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An efficient and incentive-compatible market design for energy storage participation

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HIGHLIGHTS

- A bidding structure based on cycling mileages is proposed for ESS participation.
- Clearing model is modified to adapt the bidding structure and manage SOC of ESS.
- Settlement ensures incentive compatibility, budget balance and calculation speed.
- We conduct a numerical study to illustrate advantages of the proposed mechanism.

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ABSTRACT

With the increasing penetration of renewables, energy storage systems (ESS) are becoming growingly important due to its peak-shaving ability. However, the current market mechanism is not well prepared for the participation of the ESSs. Firstly, the current bidding structure requires the ESSs to submit separate parameters for charging and discharging, but this structure is inconsistent with their operating characteristics. Secondly, the current settlement rule settles the ESSs according to time-variant locational marginal prices (LMP), but the diminishing intertemporal price spreads will encourage them to behave strategically. To this end, this paper proposes a novel bidding structure, a corresponding clearing model and a modified settlement rule: The bidding structure for the ESSs includes cost functions with respect to cycling mileages and valuation functions for ending stored energy. Thereafter the independent system operators (ISO) will manage the ESSs' state of charge (SOC) and clear the market. A settlement rule based on Vickery-Clarke-Groves (VCG) mechanism and Asymmetric Nash Bargaining theory is adopted to incentivize the ESSs to behave honestly. Numerical tests are conducted to illustrate the social welfare efficiency, incentive compatibility and computational tractability of the proposed mechanism.

1. Introduction

Energy storage systems (ESS) are considered as a promising solution to the challenges brought by the increasing penetration of renewables. They can store surplus energy during off-peak hours and discharge it during peak hours, so as to contribute to renewable power consumption and system adequacy [1,2].

To promote sustainable development of the ESSs, their remuneration mechanism needs to be carefully designed. In the context of market deregulation, the ESSs with growing scales are expected to act as active market participants, make profits and recover investment costs in the market.

There is a growing number of literatures that analyze ESSs participation in the market from a game-theoretic perspective. Some researchers analyze the optimal bidding strategies of ESS owners. Reference [3] models virtual energy storages as price-makers of the market, and formulates a bi-level stochastic model to derive its optimal bidding. Reference [4] obtains the optimal strategy of a merchant storage owner with unilateral market power. Reference [5] jointly considers investment and operating strategies for merchant ESS owners to gain more profit. Some researchers analyze the competition equilibrium of the market. In reference [6], an equilibrium model with equilibrium constraints is formulated to analyze the gaming behaviors among multiple market participants. The equilibrium results are derived

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for a master-slave game between an electricity system operator and a district heat storage provider in reference [7]. Reference [8] constructs a Stackelberg model to analyze the coordinated ESS investment from merchant owners and social planner. These works have made great contributions and provide inspirations for either market participants or analyzers.

Apart from them, from the perspective of market designers, the mechanism also needs to be carefully designed to accommodate the ESS participation. Generally, there are two mechanism modes for market participation of the ESSs. The first mode is to establish an independently operated market for storage usage rights, where generators or consumers can purchase usage rights from storage owners by centralized auction [9], collective sharing [10,11], bilateral contracting [12] or peer-to-peer trading [13]. This mode operates easily, but the decoupled operation between storage usage rights markets and energy markets cannot guarantee the optimal allocation of the storage resources. The second mode is to directly integrate the ESSs by the current pool-based electricity market, where the clearing model will jointly dispatch all the resources and determine the welfare-maximizing clearing result. This mode can reach a global optimum in theory, so it is embraced by the markets in North America. In February 2018, Federal Energy Regulatory Commission (FERC) issued Order 841 [14], requiring the independent system operators (ISO) to facilitate direct market participation of the ESSs.

After the issue of Order 841, several ISOs in the United States have drafted reform plans. For example, PJM defines a new resource model named Electric Storage Resource (ESR) [15]. Under this model, the ESSs can submit their cost curves for energy provision and utility functions for energy consumption. Based on the collected parameters, PJM will clear the market, determine the scheduling plans and generate locational marginal prices (LMP) [16,17].

The new model takes a big step forward, but it still has two weaknesses that might hinder the effective utilization of the ESSs: First, the bidding structure requires the ESSs to submit separated charging utility functions and discharging cost curves, but this is inconsistent with the coupled relationship between charging and discharging [18]. Secondly, the energy exchanged with the market are settled at the time-variant LMP, but this might distort their incentives to behave honestly [19,20].

The current bidding structure cannot help the ESSs to suggest their accurate operating parameters. The ESSs will be confused by the calculation of their discharging costs and charging utilities, since they actually interdepend on each other [21]. For example, the ESSs' discharging cost curves depend heavily on the energy prices they pay for charging. However, the charging prices that are generated in the day-ahead clearing are unknown to the ESSs when they bid. In fact, the main costs for the ESSs, batteries in particular, come from cycling degradation costs that grow with cycling mileages [22]. Therefore, a new bidding structure based on cycling mileages and charging-discharging pairs should be established.

The current settlement rule may discourage the ESSs from behaving honestly. Now the ESSs profit by arbitraging intertemporal price spreads, and the price spreads will diminish as they perform peak-shaving. The diminishing spreads contribute to larger social welfare but might undermine the ESSs' profit [23]. To resolve the conflict between collective interests and individual interests, a new mechanism should be introduced that settles the ESSs based on their welfare contribution.

Some researches have already explored the effects of bidding structures. In reference [24,25], different bidding structures are compared in terms of the realized storage profit and social welfare. Some bidding structures allow the ESSs to de-rate their capacities and submit time-variant maximal output, and some bidding structures allow the ESSs to simultaneously submit charging and discharging economic bids. These novel structures provide more flexibility in the bidding and contribute to greater storage profit and social welfare, but they do not consider that the variable degradation costs are functions with respect to cycling mileages. Furthermore, these researches do not develop the corresponding clearing model that manages the coupled relationship between charging and discharging.

Some researches have already introduced Vickery-Clarke-Groves (VCG) [26], a mechanism that settles an agent based on its contribution, into the electricity market. It can successfully mitigate strategic behaviors of generators or consumers [27,28]. However, implementing the classical VCG mechanism directly on the ESSs might be computationally demanding. When determining the settlement to each agent, VCG mechanism needs to clear the market once again with the agent excluded in order to calculate its marginal welfare contribution [29]. Considering the large number of the distributed ESSs, the mechanism needs proper simplifications to be computationally tractable. Additionally, the traditional VCG mechanism will incur budget deficit [30], so rules need to be designed to allocate the deficit fairly.

To adapt to the characteristics of the ESSs, this paper proposes a modified market framework. The ESSs will submit their degradation costs per cycling mileage. Since the initial and ending state of charge (SOC) can be different in the day-ahead market [31,32], the ESSs are also allowed to suggest their valuation for the remaining energy. To match the bidding structure, a corresponding market clearing model will consider the bidding information and manage the SOC constraints. As for the settlement rule, we adopt a mechanism combining VCG mechanism and Asymmetric Nash Bargaining theory. The aggregated payment to all the ESSs is determined first and then allocated among the ESSs based on Asymmetric Nash Bargaining theory [33,34]. To ensure budget balance, the deficit is properly allocated among other market participants according to the beneficiary principle. To highlight our contributions, in this paper we focus on the VCG application to the ESSs. It should be pointed out that the VCG settlement on the generators and consumers proposed in reference [27,28] can be applied on top of our mechanism easily.

The contribution of the paper could be summarized as follows:

(1) A novel bidding structure based on cycling mileages is proposed to adapt to the coupled relationship between charging and discharging of the ESSs. Besides, the ESSs can suggest their valuation functions for the ending SOC of the day-ahead market. The current bidding structure fails to allow the ESSs to suggest their accurate operating characteristics, and the proposed structure fixes the problem.
(2) A corresponding clearing model that matches the bidding structure is developed. The cycling costs and ending SOC valuation functions are modelled in the objective function, and the SOC constraints are added in the model. In contrast, the current clearing model does not incorporate these costs and constraints.
(3) This paper proposes an incentive compatible and computationally efficient settlement rule for the ESSs. The VCG mechanism is used first to derive the aggregated payment to all the ESSs based on their contribution, and Asymmetric Nash Bargaining theory is then used to allocate the payment. The budget deficit is properly allocated among the market participants. The proposed can provide better incentives for the ESSs to behave honestly compared with the current rule.

The rest of the paper is organized as follows. Section 2 presents the overall market framework. In Section 3, we illustrate the proposed bidding structure and the corresponding clearing model. Section 4 demonstrates the settlement rule design. Section 5 conducts a numerical study to verify the effectiveness of the mechanism. Section 6 concludes the paper.

2. Framework of the proposed mechanism

The proposed mechanism is an extension of the pool-based spot market that is widely implemented in North America [35–37], with

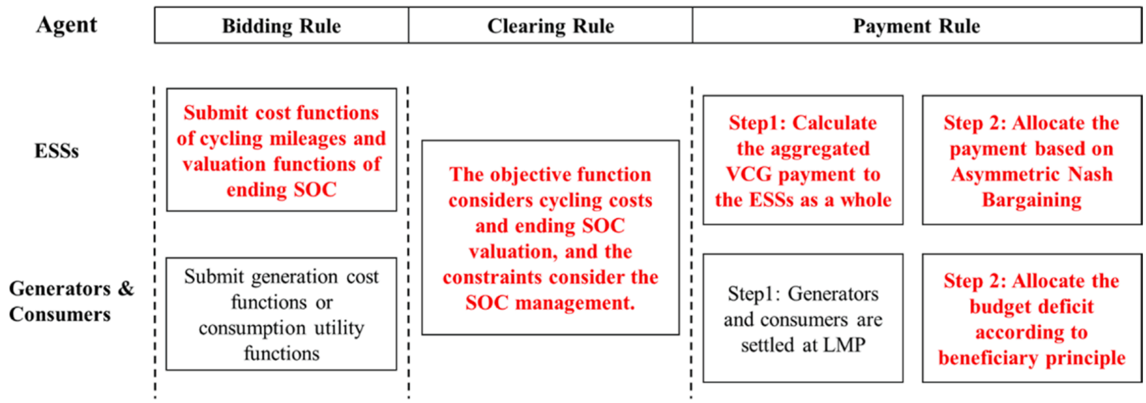


Fig. 1. Flow chart of the proposed mechanism.

several modifications designed for the ESSs. The flow chart of the mechanism is illustrated in Fig. 1. It consists of the bidding rule, the clearing rule and the payment rule. The modifications are highlighted in bold red font in Fig. 1.

In the market organization streamline, the market participants firstly submit their parameters to the ISO. Following the current market practice, generators and consumers still submit generation cost functions or consumption utility functions. As for the ESSs, they will submit their cost functions with respect to cycling mileages and valuation functions for the ending SOC.

After the ISO collect all the bids from the market participants, it will clear the market. Specifically, cycling costs and ending SOC valuation are considered in the objective function, and the SOC constraints are modelled in the clearing model. Aiming to maximize the social welfare, the market clearing will determine the generation output, the consumer consumption, the scheduling plans of the ESSs and the network power flow.

Thereafter, the settlements to the market participants are calculated. The aggregated VCG payment to all the ESSs is determined first based on their marginal market contribution, and then the payment is allocated among all the ESSs by using Asymmetric Nash Bargaining theory. The generators and consumers will still be settled by LMP, but those who benefit from storage participation will pay to cover the market budget deficit incurred by the VCG settlement to the ESSs.

3. Bidding rule and clearing model

This section introduces the modified bidding structure where the ESSs will submit cost per cycling mileage and ending SOC valuation. In the corresponding clearing model, the ISO will consider the submitted parameters in the objective function and the SOC constraints in the clearing.

3.1. Bidding rule

3.1.1. Bidding structure for the ESSs

Unlike the generators whose operating costs grow with generation power, the ESSs have operating costs that grow with cycling mileages. According to reference [38], the main variable costs for the ESSs are degradation costs that arise from the gradually aging in the cycling process. Therefore, it is reasonable to allow them to suggest their degradation costs per cycling mileage. Reference [39] further finds that the aging rate grows exponentially with the charging or discharging power rate, so the marginal degradation costs also increases with the power exchange rate. For computational tractability, the bidding structure is devised to be a stepwise function so that the optimization problem remains linear. For ESS i, it can submit K blocks for its marginal degradation cost. $c_{i,k}^{\mathrm{ESS,ch(dis)}}$ represents its marginal degradation cost for block k, and $\overline{P_{i,k}^{\mathrm{ESS,ch(dis)}}}$ represents the length of the block k for the charging or discharging power rate.

Some researches design the market clearing model based on the prerequisite that the ending SOC of the market time horizon equals the initial SOC [40,41]. In fact, some energy can be stored for future discharge if the current market prices are not profitable, or some energy can be used if the current prices are lucrative enough [31,32]. Therefore, the proposed mechanism allows the ESSs to suggest their valuation function for the ending SOC, and the marginal valuation function is also designed to be in a stepwise format. For ESS i, $c_{i,k}^{\mathrm{ESS,SOCend}}$ represents the marginal ending SOC valuation for block k, and $\overline{SOC_{i,k}^{\mathrm{ESS,end}}}$ represents the length of the block k for the ending SOC.

Additionally, ESS i will submit its maximal charging and discharging power rate $\overline{P_i^{\mathrm{ESS,ch}}}, \overline{P_i^{\mathrm{ESS,dis}}}$, maximal and minimal state of charge $\overline{SOC_i^{\mathrm{ESS}}}$, $\underline{SOC_i^{\mathrm{ESS}}}$, charging and discharging efficiency $\eta_i^{\mathrm{ESS,ch}}, \eta_i^{\mathrm{ESS,dis}}$, energy capacity $E_i^{\mathrm{ESS,cap}}$.

3.1.2. Bidding structure for generators and consumers

The generators and consumers will still submit the same parameters as they do in the current pool-based market. Generator j submits its generation cost function in a stepwise format, where $c_{j,k,t}^{\mathrm{G}}$ and $\overline{P_{j,k,t}^{\mathrm{G}}}$ denote its marginal generation cost for block k and the length of the corresponding block at time slot t. As a function of the generation output, the generation cost is affected by the fuel cost, maintainence cost and degradation cost.

The consumer k submits its consumption utility function, where $c_{d,k,t}^{\mathrm{D}}$ and $\overline{P_{d,k,t}^{\mathrm{D}}}$ denote its marginal consumption utility and the length of the block correspondingly. As a function of the energy consumption, the utility is affected by the user's willingness to pay.

Note that the bidding function could be time-variant, which is consistent with the current market rule in the electricity markets in North America [36].

3.2. Clearing rule

According to the bidding parameters, the ISO clears the market by solving the following optimization problem. For simplicity, the generator unit commitment is disregarded in this part. The decision variables are the generation output $P_{j,k,t}^{\mathrm{G}}$, the consumer consumption $P_{d,k,t}^{\mathrm{D}}$, the operation schedule of the ESSs $P_{i,k,t}^{\mathrm{ESS,ch}}, P_{i,k,t}^{\mathrm{ESS,dis}}$, the SOC of the ESSs $SOC_{i,t}^{\mathrm{ESS}}$ and the phase angle of each node $\delta_{n,t}$. Besides, the ending SOC of the ESSs $SOC_{i,k}^{\mathrm{ESS,end}}$ are also decision variables. The subscript i, j, d, k, t represents the index of the ESSs, generators, consumers, blocks and time slots respectively.

The objective function (1.1) is to maximize total social welfare. The first item represents the aggregated consumer utilities and the second item represents the aggregated generation costs. The third item represents the cycling degradation costs of the ESSs, and the fourth item represents the value of the remaining energy in the ESSs.

$$maxS = \left\{ \begin{array}{c} \sum_{d,k,t} c^{\mathrm{D}}_{d,k,t} P^{\mathrm{D}}_{d,k,t} - \sum_{j,k,t} c^{\mathrm{G}}_{j,k,t} P^{\mathrm{G}}_{j,k,t} - \\ \sum_{i \in I,k,t} \left(c^{\mathrm{ESS,dis}}_{i,k} P^{\mathrm{ESS,dis}}_{i,k,t} + c^{\mathrm{ESS,ch}}_{i,k} P^{\mathrm{ESS,ch}}_{i,k,t} \right) + \sum_{i \in I,k} \left(c^{\mathrm{ESS,SOCend}}_{i,k} \mathrm{SOC}^{\mathrm{ESS,end}}_{i,k} \right) \end{array} \right\} \tag{1.1}$$

The constraints are as follows, where I, J, D represents the system-wide sets for ESSs, generators and consumers. $\Omega^{\mathrm{G}}_n, \Omega^{\mathrm{D}}_n, \Omega^{\mathrm{ESS}}_n$ represents the sets of generators, consumers and ESSs at node n.

$$0 \leqslant P^{\mathrm{D}}_{d,k,t} \leqslant \overline{P^{\mathrm{D}}_{d,k,t}} : \underline{\mu^{\mathrm{D}}_{d,k,t}}, \overline{\mu^{\mathrm{D}}_{d,k,t}}, \forall d \in D, k, t \tag{1.2}$$

$$0 \leqslant P^{\mathrm{G}}_{j,k,t} \leqslant \overline{P^{\mathrm{G}}_{j,k,t}} : \underline{\mu^{\mathrm{G}}_{j,k,t}}, \overline{\mu^{\mathrm{G}}_{j,k,t}}, \forall j \in J, k, t \tag{1.3}$$

$$0 \leqslant P^{\mathrm{ESS,ch}}_{i,k,t} \leqslant \overline{P^{\mathrm{ESS,ch}}_{i,k}} : \underline{\mu^{\mathrm{ESS,ch}}_{i,k,t}}, \overline{\mu^{\mathrm{ESS,ch}}_{i,k,t}}, \forall i \in I, k, t \tag{1.4}$$

$$0 \leqslant P^{\mathrm{ESS,dis}}_{i,k,t} \leqslant \overline{P^{\mathrm{ESS,dis}}_{i,k}} : \underline{\mu^{\mathrm{ESS,dis}}_{i,k,t}}, \overline{\mu^{\mathrm{ESS,dis}}_{i,k,t}}, \forall i \in I, k, t \tag{1.5}$$

$$\mathrm{SOC}^{\mathrm{ESS}}_{i,t} E^{\mathrm{ESS,cap}}_i = \mathrm{SOC}^{\mathrm{ESS}}_{i,t-1} E^{\mathrm{ESS,cap}}_i + \eta^{\mathrm{ESS,ch}}_i \sum_k P^{\mathrm{ESS,ch}}_{i,k,t} - \frac{\sum_k P^{\mathrm{ESS,dis}}_{i,k,t}}{\eta^{\mathrm{ESS,dis}}_i}, \forall i \in I, t \geqslant 2 \tag{1.6}$$

$$\underline{\mathrm{SOC}^{\mathrm{ESS}}_i} E^{\mathrm{ESS,cap}}_i \leqslant \mathrm{SOC}^{\mathrm{ESS}}_{i,t} E^{\mathrm{ESS,cap}}_i \leqslant \overline{\mathrm{SOC}^{\mathrm{ESS}}_i} E^{\mathrm{ESS,cap}}_i : \underline{\tau^{\mathrm{ESS}}_{i,t}}, \overline{\tau^{\mathrm{ESS}}_{i,t}}, \forall i \in I, t \tag{1.7}$$

$$\left\{ \begin{array}{c} \sum_{j \in \Omega^{\mathrm{G}}_n} P^{\mathrm{G}}_{j,t} + \sum_{i \in \Omega^{\mathrm{ESS}}_n} \left[P^{\mathrm{ESS,dis}}_{i,t} - P^{\mathrm{ESS,cha}}_{i,t} \right] \\ - \sum_{d \in \Omega^{\mathrm{D}}_n} P^{\mathrm{D}}_{d,t} - \sum_{m \in \psi_n} B_{nm} \left(\delta_{n,t} - \delta_{m,t} \right) \end{array} \right\} = 0 : \lambda_{n,t}, \forall n, t \tag{1.8}$$

$$B_{nm} \left(\delta_{n,t} - \delta_{m,t} \right) \leqslant \overline{P^{\mathrm{L}}_{nm}} : \pi_{nm,t}, \forall n, \forall m \in \psi_n \tag{1.9}$$

$$\delta_{\mathrm{refn},t} = 0 \tag{1.10}$$

$$E^{\mathrm{ESS,cap}}_i \left(\mathrm{SOC}^{\mathrm{ESS}}_{i,T} - \mathrm{SOC}^{\mathrm{ESS}}_{i,0} \right) = E^{\mathrm{ESS,cap}}_i \sum_k \mathrm{SOC}^{\mathrm{ESS,end}}_{i,k} : \nu^{\mathrm{ESS,end}}_i, \forall i \in I \tag{1.11}$$

$$0 \leqslant \mathrm{SOC}^{\mathrm{ESS,end}}_{i,k} \leqslant \overline{\mathrm{SOC}^{\mathrm{ESS,end}}_{i,k}} : \underline{\mu^{\mathrm{ESS,endSOC}}_{i,k}}, \overline{\mu^{\mathrm{ESS,endSOC}}_{i,k}}, \forall i \in I, k \tag{1.12}$$

$$\sum_k P^{\mathrm{D}}_{d,k,t} = P^{\mathrm{D}}_{d,t}, \forall d \in D, t \tag{1.13}$$

$$\sum_k P^{\mathrm{G}}_{j,k,t} = P^{\mathrm{G}}_{j,t}, \forall j \in J, t \tag{1.14}$$

$$\sum_k P^{\mathrm{ESS,dis(ch)}}_{i,k,t} = P^{\mathrm{ESS,dis(ch)}}_{i,t}, \forall i \in I, t \tag{1.15}$$

Constraint (1.2) represents the maximal and minimal consumer power consumption. Constraint (1.3) represents the maximal and minimal generation output. Constraint (1.4)(1.5) represent the constraints on charging rates and discharging rates for ESS i. Constraint (1.6) calculates the SOC for ESS i. Constraint (1.7) represents the maximal and minimal SOC constraint for ESS i. The nodal power balance is illustrated in (1.8) and the capacity constraint of the transmission line is represented in (1.9). Constraint (1.10) defines the reference node. Constraint (1.11) calculates the ending SOC for each block and constraint (1.12) determines that the ending SOC for each block cannot exceed the block length. Equation (1.13)-(1.15) formulates the relationship between the aggregated output/consumption/charge/discharge and the values for each block. The Lagrange multipliers are listed after colons of the constraints. Specially, $\lambda_{n,t}$ can be seen as the LMP at node n at time slot t.

This model actually manages the SOC of the ESSs, and considers their cycling costs and ending SOC valuation in the objective function. As a good match of the bidding structure, the clearing model can maximize the social welfare and ensure physical feasibility of the ESS scheduling plan. After solving the optimization problem, the ISO gets the clearing result, including the generator output $\mathbf{P}^{\mathrm{G},*}_{\mathbf{J}}$, the consumer consumption $\mathbf{P}^{\mathrm{D},*}_{\mathbf{D}}$ and the ESS schedule $\mathbf{P}^{\mathrm{ESS},*}_{\mathbf{I}}$. The superscript * denotes the day-ahead scheduling plan that maximizes the social welfare.

4. Settlement rule

This section describes the VCG mechanism and Asymmetric Nash Bargaining theory to settle ESSs. The payment calculation of the ESSs consists of two steps. In the first step, the ESSs are viewed as a whole and their aggregated payment is calculated based on the classical VCG mechanism. In the second step, the payment is allocated among the ESSs based on Asymmetric Nash Bargaining rule. Since the VCG payment to the ESSs causes budget deficit, the deficit is allocated among the generators and consumers according to the beneficiary rule.

It is notable that the VCG-class mechanism can also be applied to generators or consumes. The derivation of settlement rule will be different since they have different kinds of strategic behaviors, like economic or physical withholdings for generators [42,43]. Since the core issue of the paper is ESS market participation, we do not design and derive the VCG-class settlement rule for other market participants, and the corresponding work has been covered in reference [27,28].

4.1. Step 1: Aggregated payment to ESSs based on VCG mechanism

In the VCG mechanism, the payment made to an agent is determined by its marginal contribution. The contribution is defined as the improvement of other participants' welfare after the bidder's participation [26]. Therefore, we calculate the welfare differences between the following two cases: (1) the welfare with the corresponding bidder scheduled at the day-ahead clearing results. (2) the welfare with the corresponding bidder excluded from the market.

We define a function S_{-I} to denote the maximal welfare of other market participants that can be obtained with a certain scheduling plan of the ESSs. The argument $\mathbf{c}_{-\mathbf{I}}$ denotes the submitted bids excluding the ESSs, so only bids of the consumers and generators are included. The argument $\mathbf{P}^{\mathbf{ESS}}_{\mathbf{I}}$ is the schedule of the ESSs. The function is defined as follows:

$$S_{-I}(\boldsymbol{c}_{-I}, \boldsymbol{P}^{ESS}_{I}) = \max_{\boldsymbol{P}^{D}_{D}, \boldsymbol{P}^{G}_{J}, \delta} \left[\sum_{d,k,t} c^{\mathrm{D}}_{d,k,t} P^{\mathrm{D}}_{d,k,t} - \sum_{j,k,t} c^{\mathrm{G}}_{j,k,t} P^{\mathrm{G}}_{j,k,t} \right] \tag{2.1}$$

$$s.t(1.2)(1.3)(1.8)(1.9)(1.10)(1.13)(1.14) \tag{2.2}$$

In the above function, when $\mathbf{P}^{\mathbf{ESS}}_{\mathbf{I}}$ equals 0, it means that the ESSs are excluded from the day-ahead market. When $\mathbf{P}^{\mathbf{ESS}}_{\mathbf{I}}$ equals $\mathbf{P}^{\mathbf{ESS},*}_{\mathbf{I}}$, it means the ESSs are scheduled optimally according to the day-ahead clearing results.

The VCG payment made to the ESSs can be expressed as:

$$R^{ESS,VCG} = S_{-I}(c^0_{-I}, P^{ESS*}_I) - S_{-I}(c^0_{-I}, 0) \tag{2.3}$$

$S_{-I}(\mathbf{c}^0_{-\mathbf{I}}, \mathbf{P}^{\mathbf{ESS},*}_{\mathbf{I}})$ is the welfare when the ESSs are scheduled according to the day-ahead clearing results. $S_{-I}(\mathbf{c}^0_{-\mathbf{I}}, 0)$ is the welfare when the ESSs are excluded from the market. $\mathbf{c}^0_{-\mathbf{I}}$ represents the bids from generators and consumers in the day-ahead market. We assume that S_{-I} is well defined when $\mathbf{P}^{\mathbf{ESS},*}_{\mathbf{I}}$ takes the value of **0.**

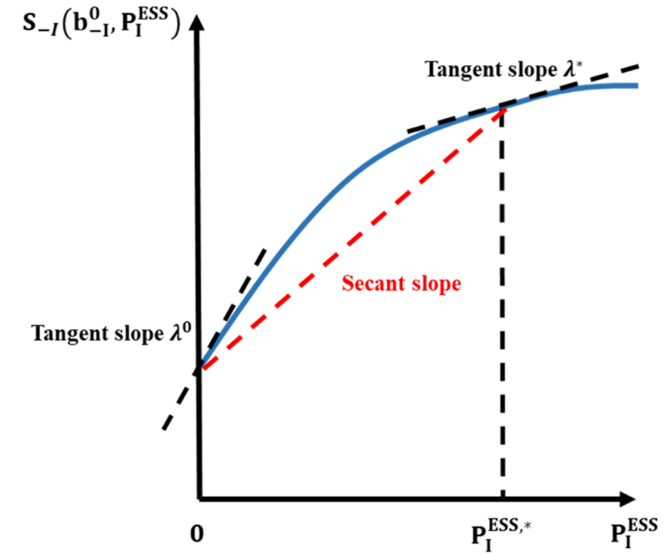


Fig. 2. Approximation of the bargaining power.

4.2. Step 2: Payment allocation among the ESSs based on Asymmetric Nash Bargaining theory

In the second step, the aggregated payment is allocated among the ESSs fairly. Asymmetric Nash Bargaining theory is introduced as an axiomatic solution to the allocation.

An Asymmetric Nash Bargaining game can be expressed as $(\mathbf{F},\mathbf{d},\alpha)$. $\mathbf{F}$ denotes the vector of allocation plan. $\mathbf{d}$ denotes the vector of disagreement point, and it constrains the minimal payment that ESSs should get. α denotes the bargaining power that reflects the importance weight factor in the allocation process.

Different from a Symmetric Nash Bargaining model, an Asymmetric Nash Bargaining problem tries to maximize the product of market participants' excess payoff with different exponents. The exponents α^i are the bargaining power of the market participants.

$$max\prod_i (F_i - d_i)^{\alpha_i} \tag{3.1}$$

$$\sum_i F_i = R^{\mathrm{ESS,VCG}} \tag{3.2}$$

$$R_i^{ESS,VCG} = F_i \geqslant d_i = R_i^{ESS,LMP} \tag{3.3}$$

Equation (3.1) is the objective function. Constraint (3.2) represents the limit of the aggregated payment that can be allocated. Constraint (3.3) represents that payment received by each agent should exceed the disagreement point.

The disagreement point d_i means the payment that each agent can get without joining the game. If ESS i refuses to join the ESS coalition, it has to participate in the market and be settled at the LMP. Therefore, the disagreement point can be obtained by calculating LMP settlement of ESS i, which is denoted by $R_i^{\mathrm{ESS,LMP}}$:

$$d_i = R_i^{\mathrm{ESS,LMP}} = \sum_t \lambda^*_{n(i\in\Omega_n^{\mathrm{ESS}}),t}\left(P_{i,t}^{\mathrm{ESS,dis},*} - P_{i,t}^{\mathrm{ESS,ch},*}\right) \tag{3.4}$$

As is proven in Appendix A, the total LMP settlement of the ESSs will be less than the VCG settlement.

$$\sum_i d_i = R^{\mathrm{ESS,LMP}} \leqslant R^{\mathrm{ESS,VCG}} \tag{3.5}$$

Therefore, we can conclude that there must be feasible solutions in the feasible region constrained by (3.2) and (3.3).

We can calculate the bargaining power α_i by using approximative market contribution. The contribution of the ESS is the welfare increase due to its power delivery from peak hours to valley hours. Since the LMP is the welfare disturbance due to per unit disturbance in nodal power injection, we can approximate the ESS's contribution by averaging the LMPs under two cases: the market result with the ESSs optimally scheduled, the market result with all the ESSs excluded.

We use $(\mathbf{P}_\mathbf{D}^{\mathbf{D},*}, \mathbf{P}_\mathbf{J}^{\mathbf{G},*}, \delta^*, \mathbf{P}_\mathbf{I}^{\mathbf{ESS},*})$ to denote the market clearing results with the ESSs optimally scheduled, and use λ^* to denote the corresponding LMP. We denote the scheduling plan with all the ESSs excluded as $(\mathbf{P}_\mathbf{K}^{\mathbf{D},0}, \mathbf{P}_\mathbf{J}^{\mathbf{G},0}, \delta^0, 0)$ and the corresponding LMP as λ^0. At node n and time slot t, the changes in welfare due to per megawatt energy injected can be approximated as $\frac{\lambda^*_{n,t}+\lambda^0_{n,t}}{2}$. An intuitive demonstration is shown in Fig. 2: The secant slope can be approximated by averaging the two tangent slopes at the endpoints.

Therefore, the bargaining power can be denoted as follows:

$$\alpha_i = \sum_t \frac{\lambda^*_{n(i\in\Omega_n^{\mathrm{ESS}}),t} + \lambda^0_{n(i\in\Omega_n^{\mathrm{ESS}}),t}}{2}\left(P_{i,t}^{\mathrm{ESS,dis},*} - P_{i,t}^{\mathrm{ESS,ch},*}\right) \tag{3.6}$$

After proper reformulation, the solution to the optimization problem (3.1)-(3.3) can be easily derived by using Karush-Kuhn-Tucker (KKT) theorem [44,45]. The detailed formulation is omitted due to limited space. $R_i^{\mathrm{ESS,VCG}}$ denotes the payment to ESS i under the proposed mechanism.

$$R_i^{\mathrm{ESS,VCG}} = F_i = R_i^{\mathrm{ESS,LMP}} + \left(R^{\mathrm{ESS,VCG}} - R^{\mathrm{ESS,LMP}}\right)\frac{\alpha_i}{\sum_k \alpha_k} \tag{3.7}$$

4.3. Satisfaction of preferable properties

The proposed settlement rule can satisfy the following preferable properties:

(a) **Pareto efficiency**: It denotes a situation that cannot be modified to make any individual better off without undermining at least one agent's interest. If there exists a new reward vector $\overrightarrow{\mathbf{R}^{\mathbf{ESS,VCG}}}$ that satisfies $\overrightarrow{R_i^{\mathrm{ESS,VCG}}} \geq R_i^{\mathrm{ESS,VCG}}$ for any i (and the inequality holds for at least one element), we have

$$\sum_i \overrightarrow{R_i^{\mathrm{ESS,VCG}}} > \sum_i R_i^{\mathrm{ESS,VCG}} = R^{\mathrm{ESS,VCG}} \tag{3.8}$$

The allocated payments to the ESSs will exceed the total budget $R^{\mathrm{ESS,VCG}}$. Therefore, there does not exist any allocation where one can earn more without grasping others' surplus, and *Pareto efficiency* holds.

(b) **Monotonicity**: It denotes an allocation plan where the agent who makes more contribution always gets higher payoffs. For any i,j that satisfy $\alpha_i > \alpha_j$, we have

$$\begin{aligned} R_i^{\mathrm{ESS,VCG}} &= R_i^{\mathrm{ESS,LMP}} + \left(R^{\mathrm{ESS,VCG}} - R^{\mathrm{ESS,LMP}}\right)\frac{\alpha_i}{\sum_k \alpha_k} \\ &> R_j^{\mathrm{ESS,LMP}} + \left(R^{\mathrm{ESS,VCG}} - R^{\mathrm{ESS,LMP}}\right)\frac{\alpha_j}{\sum_k \alpha_k} = R_j^{\mathrm{ESS,VCG}} \end{aligned} \tag{3.9}$$

The more the ESS contributes to the welfare, the bigger the bargaining power α_i and the higher the payoff $R_i^{\mathrm{ESS,VCG}}$. Therefore, *monotonicity* holds.

(c) **Individual rationality**: It requires that any agent is better off by participating in the proposed mechanism than not. Firstly, according to (3.3), we can conclude that the allocation determined by Asymmetric Nash Bargaining ($R_i^{\mathrm{ESS,VCG}}$) is no smaller than $R_i^{\mathrm{ESS,LMP}}$. Besides, it could be proved that the LMP settlement $R_i^{\mathrm{ESS,LMP}}$ can always cover the costs of the ESSs (see Appendix B). Therefore, the following equation holds.

$$R_i^{\mathrm{ESS,VCG}} \geqslant R_i^{\mathrm{ESS,LMP}} \geqslant C_i^{\mathrm{ESS}} \tag{3.10}$$

Table 1
Technical Parameters of the ESSs.

Index	Maximal discharge/charge rate	EnergyCapacity	Initial SOC	Cycling Costs per mileage	Charging and discharging efficiency	Ending SOC valuation
ESS 1	80 MW	160MWh	0.5	3–4$/MWh	90%	35–55$/MWh
ESS 2	100 MW	200MWh	0.5	4–6$/MWh	90%	35–55$/MWh
ESS 3	100 MW	200MWh	0.5	1–1.5$/MWh	85%	35–55$/MWh
ESS 4	200 MW	400MWh	0.5	5–8$/MWh	92%	35–55$/MWh
ESS 5	160 MW	320MWh	0.5	3–4$/MWh	90%	35–55$/MWh
ESS 6	100 MW	200MWh	0.5	4–6$/MWh	90%	35–55$/MWh
ESS 7	100 MW	200MWh	0.5	1–1.5$/MWh	85%	35–55$/MWh
ESS 8	200 MW	400MWh	0.5	5–8$/MWh	92%	35–55$/MWh

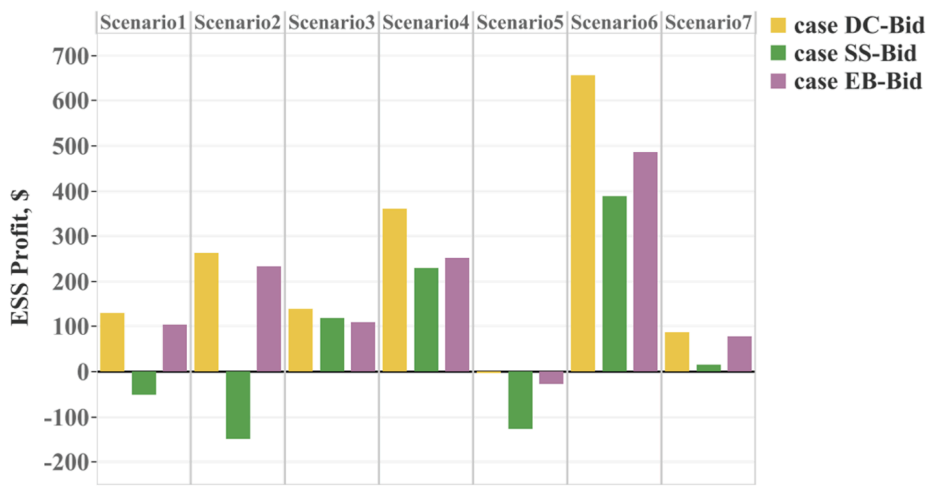


Fig. 3. ESS Profit comparison of different bidding structures.

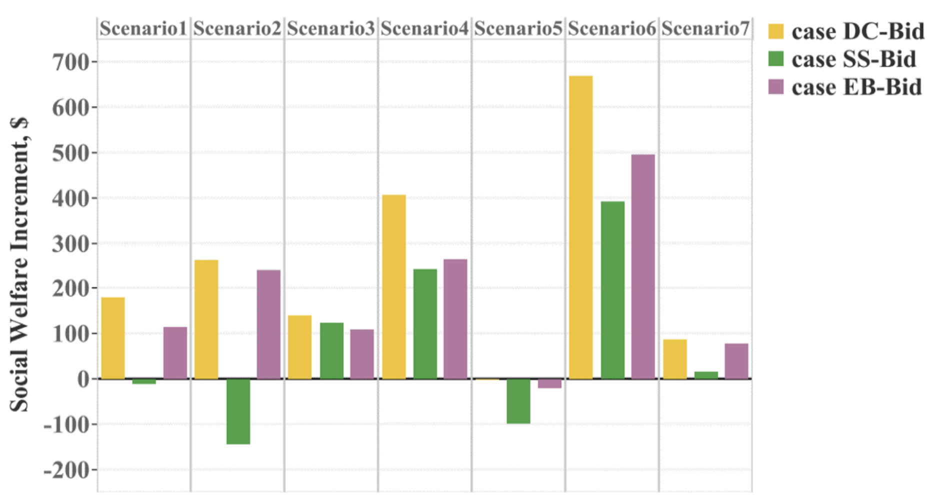
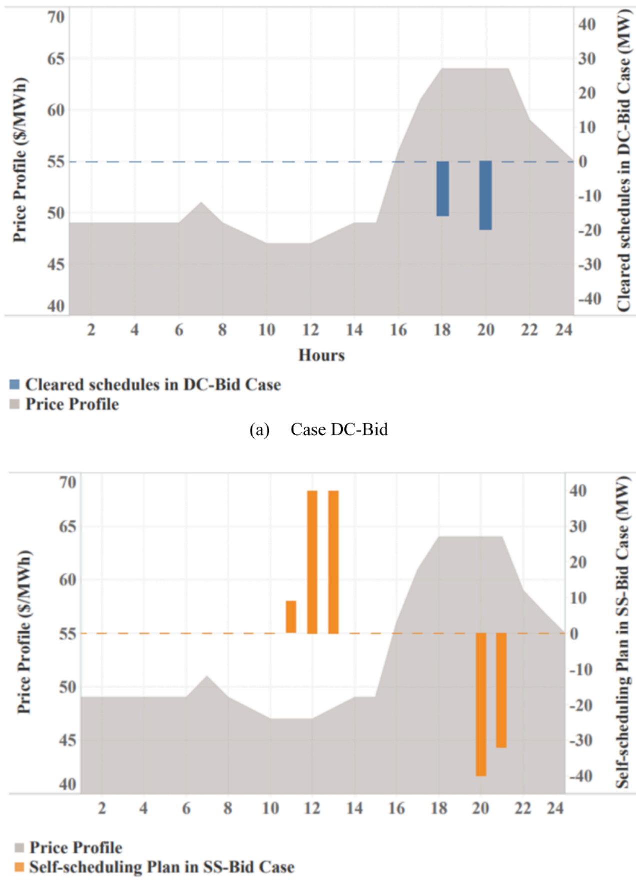


Fig. 4. Social welfare increment of different bidding structures.

This means that it is best for the ESSs to get involved in the VCG settlement, and participating in the traditional LMP settlement takes the second place, and it is worst for the ESSs to exit the market. Therefore, *individual rationality* holds.

(d) **Incentive compatibility:** When the ownership of the ESSs is distributed, a single ESS cannot affect prices significantly. Since *monotonicity* holds between its contribution and reward, submitting its true parameters to make more contributions is an acceptable strategy. When the ownership of the ESSs is concentrated, the monopolistic ESS will gain the highest utility when submitting its true parameters, just like colluded small agents. The proof is shown in Appendix C.

(e) **Anti-collusion:** When the ESSs form a grand coalition to submit inflated cycling degradation costs, the coalition can be seen as one single agent and the total payment received is determined by VCG mechanism in Step 1. Since the aggregated payment to the ESSs is determined as a whole first no matter the number of ESS owners, there is no difference between the payment received by a monopolistic ESS owner and a grand coalition formed by several ESS owners. As is shown in Appendix C, the coalition can be represented by a monopolistic ESS agent, and they cannot gain extra profit by colluding to submit false parameters. Therefore, there does not exist any stable coalition that can increase each ESS's utility.



(a) Case DC-Bid

(b) Case SS-Bid

Fig. 5. Scheduling Comparison of case DC-Bid case and case SS-Bid.

4.4. Budget deficit allocation to the market participants

The VCG payment to the ESSs might cause budget deficit in the wholesale market (Appendix A), so we need to allocate the deficit to other market participants.

According to the beneficiary payment principle, we identify the market participants who benefit from the participation of the ESSs by calculating the surplus changes of generators and consumers. The generator surplus equals market incomes subtracted by generation costs. The consumer surplus equals demand utilities subtracted by market expenditures. We use ΔS_d^{D} and ΔS_j^{G} to denote surplus change of consumer d and generator j.

Thereafter, the budget deficit is allocated in proportion to the surplus changes of the market participants. The extra payment made by consumer d and generator j, $R_d^{\mathrm{D,extra}}$ and $R_j^{\mathrm{G,extra}}$, can be calculated as follows.

$$R_d^{\mathrm{D,extra}} = \frac{max\{0,\Delta S_d^{\mathrm{D}}\}}{\sum_d max\{0,\Delta S_d^{\mathrm{D}}\} + \sum_j max\{0,\Delta S_j^{\mathrm{G}}\}}\left(R^{\mathrm{ESS,VCG}} - R^{\mathrm{ESS,LMP}}\right) \tag{4.1}$$

$$R_j^{G,extra} = \frac{max\{0,\Delta S_j^{G}\}}{\sum_d max\{0,\Delta S_d^{\mathrm{D}}\} + \sum_j max\{0,\Delta S_j^{G}\}}\left(R^{ESS,VCG} - R^{ESS,LMP}\right) \tag{4.2}$$

The numerator in equation (4.1) and (4.2) represents the budget deficit that needs to be allocated. All the market participants with positive welfare changes share the deficit according to their beneficiaries.

5. Numerical example

5.1. Data description

To verify the effectiveness of the proposed mechanism, we conduct a numerical test. In the proposed system, the installed capacity is 12020 MW, with 5980 MW from thermals, 3000 MW from solar, 2000 MW from wind and 1040 MW from ESSs. Historical data from the California Independent System Operator (CAISO) [46] is used to cluster 7 representative scenarios to model the uncertainties in net load profiles. The peak net load in all scenarios is 5463 MW (scenario 2). It is notable that the uncertainties only apply to the ESSs, and the ISO actually has full knowledge of net load profiles when clearing the market. For sake of simplicity, the topology is disregarded in the current analysis.

It is assumed that the renewables have zero variable costs and the utilities of the consumers are 100$/MWh. The price biddings of thermal generators range from 20 to 73 $/MWh. The renewables, thermals and consumers are assumed to submit their true parameters to the market. The technical parameters of the ESSs are listed in Table 1.

We perform numerical experiments on a test system and make proper

Table 3
Computational time comparison of case VCG-ASB-Pay and case VCG-Pay.

Scenario	1	2	3	4	5	6	7
case VCG-ASB-Pay	86.2 s	101.6 s	90.3 s	133.3 s	148.0 s	136.4 s	137.6 s
case VCG-Pay	838.6 s	860.5 s	834.9 s	836.1 s	839.4 s	841.5 s	832.0 s

Table 2
Illustration of incentive compatibility of the proposed settlement rule.

Bidding Strategy	Profit in **case VCG-ASB-Pay** ($)	Profit in **case LMP-Pay** ($)	Profit in **case PAB-Pay** ($)	Social Welfare Contribution ($)	Intertemporal price spread ($/MWh)
Understate cycling costs by 50%	947.5	−1773.9	−4166.8	947.5	15.2
Understate cycling costs by 20%	2218.6	300.0	−961.4	2218.6	16.4
Understate cycling costs by 10%	2441.3	805.4	−319.7	2441.3	17.3
True parameters	**2442.6**	874.7	0.0	**2442.6**	17.6
Overstate cycling costs by 10%	2438.2	1004.6	310.0	2438.3	18.0
Overstate cycling costs by 20%	2377.2	1169.4	535.6	2377.2	18.4
Overstate cycling costs by 40%	2145.0	**1255.4**	726.4	2145.0	19.2
Overstate cycling costs by 60%	1867.7	1188.8	**750.1**	1867.7	19.7
Overstate cycling costs by 80%	1585.5	1091.0	674.3	1585.5	20.2
Understate endingSOC valuation by 20%	−329.4	−1897.3	−10747.0	−329.4	17.6
Understate endingSOC valuation by 10%	790.6	−777.3	−4273.3	790.6	17.6
Overstate endingSOC valuation by 10%	1302.8	360.9	0.0	1302.8	18.6
Overstate endingSOC valuation by 20%	1302.8	360.9	0.0	1302.8	18.6

Table 4
Illustration of fairness of the proposed settlement rule.

ESS index	1	2	3	4	5	6	7	8
Revenue Share in VSG-ASB-Pay	9.18%	9.10%	8.59%	18.60%	18.25%	9.09%	8.59%	18.60%
Revenue Share in VCG-Pay	9.27%	9.04%	8.53%	18.53%	18.52%	9.04%	8.53%	18.53%
Contribution Share	9.27%	9.04%	8.53%	18.53%	18.52%	9.04%	8.53%	18.53%

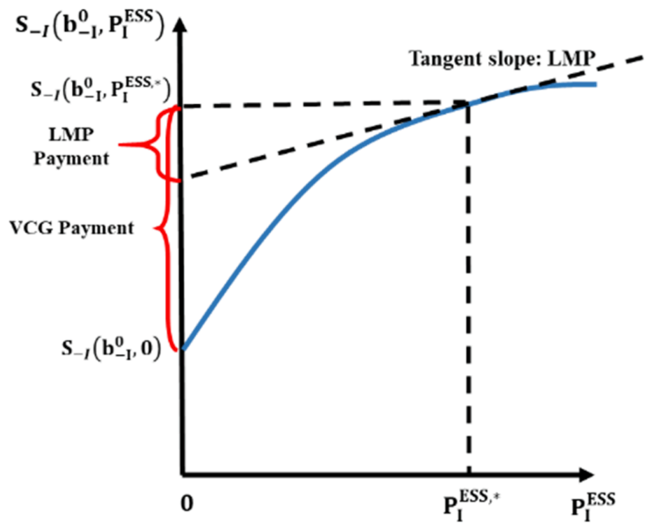


Fig. 6. Comparison between VCG payment and LMP payment.

assumptions on price biddings from market participants, since the required market bidding data for an empirical study are confidential and not available to the public. The usage of CAISO renewable output curve and demand curve can approximate the operating condition of the test system to the real system. This helps us illustrate the effects of our proposed mechanism on the test system. The market operators who have access to the confidential bidding data can make assessments on the proposed mechanism before practical implementation.

5.2. Effects of the proposed bidding structure

In this part, we illustrate the effects of the proposed bidding structure and clearing model on ESS profit and social welfare. Specifically, we design three cases: (1) **case DC-Bid**: Under the proposed bidding structure in this paper, the ESS submits its cycling degradation costs per mileage and ending SOC valuation, and the ISO considers these parameters and manages the SOC in the clearing. (2) **case SS-Bid**: The ESS guesses when the peak and off-peak hours will occur, and submit its self-scheduling plan. (3) **case EB-Bid**: The ESS guesses the clearing prices, submits economic biddings including generation cost functions and consumption utility functions, and is responsible to manage its own SOC and handle possible imbalances. **Case SS-Bid** and **case EB-Bid** serve as cases to be compared, since they represent the most prevailing two bidding mechanisms in researches [47–50] and have also been implemented in real markets such as PJM [51]. In **case SS-Bid** and **case EB-Bid**, since the ESS does not has full knowledge, it will solve a stochastic optimal bidding problem based on scenario probability distributions. In this subsection, we try to illustrate the effects of the bidding structures, so we do not introduce our proposed settlement rule and still use the time-variant LMP for settlement. ESS 1, an ESS with a small capacity (160MWh/80 MW), is chosen as the agent.

The comparison of the ESS's profit is illustrated in Fig. 3. In all the scenarios, **case DC-bid** ranks highest, with a weighted-average profit of $234.3. In **case SS-Bid**, the ESS's profit declines by 58.8% to $96.4. In case **EB-Bid**, the ESS's profit also declines by 28.5% to $167.5. Typically, in scenario 2, the ESS earns a profit as much as $262.7 in **case DC-Bid**, while it suffers a loss of $148.4 in **case SS-Bid**.

As for social welfare, **case DC-Bid** also enjoys advantage (Fig. 4). Compared with the case when all the ESSs are excluded, in **case DC-Bid** the introduction of the ESSs can contribute to a weighted-average welfare increment of $244.5. In comparison, the social welfare increment is only $110.6 in **case SS-Bid** and $173.9 in **case EB-Bid**. Typically, in scenario 2, the introduction of the ESS decreases the social welfare by $144.0 in **case SS-Bid**. In contrast, the ESS scheduling is significantly more efficient in **case DC-Bid**, with a social welfare increment of $262.7.

To show how the profit loss and welfare loss are incurred in **case SS-Bid**, the ESS scheduling plan and the price profile for scenario 2 is plotted in Fig. 5. In this scenario, the net load is relatively high, and this leaves the price relatively high accordingly. The price range (47–64 $/MWh) is higher than the ESS's valuation for stored energy. Therefore, the optimal scheduling plan is to use the stored energy for discharging, and this is consistent with the market clearing result in **case DC-Bid**. In contrast, in **case SS-Bid**, since the ESS has predicted the high net load profile, it self-schedules to maintain the SOC level and this will lower its utility. Additionally, the charging hours (11:00–14:00) mismatch the price valley hours (10:00–12:00) and the discharging hours (20:00–22:00) mismatch the price peak hours (18:00–21:00), so its profit is further reduced. The suboptimal scheduling of the ESS will also undermine the social welfare, because it cannot efficiently replace peak generators and save fuel costs.

It can be concluded that, the proposed bidding structure can overcome the weakness underlying the current bidding schedule. In the proposed mechanism (**case DC-Bid**), the ESS can effectively suggest its variable cycling costs per mileage and ending SOC valuation, and the ISO will manage the SOC flexibly according to different net load scenarios. In contrast, in **case SS-Bid** and **case EB-Bid,** the ESS has to take into account all the possible scenarios in their bidding. Therefore, the proposed mechanism can increase the ESSs' profit and the social welfare.

5.3. Effects of the proposed settlement rule

We compare the settlement under the proposed mechanism with the current LMP mechanism, and the Pay-as-bid (PAB) mechanism and classical VCG mechanism discussed in the literature. Four cases are considered: (1) In the proposed mechanism, the aggregated payment is determined by the VCG mechanism first, and the payment is further allocated among the ESSs by Asymmetric Nash Bargaining theory (**case VCG-ASB-Pay**). (2) All the energy exchanged between the ESSs and the market are settled by the LMP (**case LMP-Pay**). (3) All the ESSs are settled by their price biddings (**case PAB-Pay).** (4) All the ESSs are settled by the classical VCG mechanism (**case VCG-Pay**).

The last three cases serve as cases to be compared. **Case LMP-Pay** represents mechanism that has been already implemented in the current market [51], while **case PAB-Pay** and **case VCG-Pay** represent two prevailing settlement rules that have been widely discussed in researches [27,28,52,53]. In the above cases, we try to illustrate the effects of the settlement rule, so we all use the proposed bidding structure and clearing model in this paper.

5.4. Satisfaction of incentive compatibility

The comparison of the ESSs' aggregated profit in **case VCG-ASB-**

Pay, case LMP-Pay and **case PAB-Pay** is shown in Table 2. To illustrate the *incentive compatibility* of the proposed settlement, we calculate the profit under different bidding strategies.Table 3

It can be seen that in **case LMP-Pay** gaming about their parameters may be profitable. When the ESSs overstate their cycling degradation costs by 40%, the intertemporal price spread will be manipulated to a higher level (19.2 $/MWh) than that when they submit true parameters (17.6$/MWh). Therefore, the ESSs will earn $1255.4, and this is 43.5% higher than their profit when being honest.

In **case PAB-Pay**, the ESSs will also game about their parameters. When they submit true parameters, since their payment are determined by their price biddings, their income will exactly equal their costs and the corresponding profit margin will be zero. When they exaggerate cycling costs by 60%, their income will exceed their costs so they gain profit of $750.1.

In our proposed mechanism (**case VCG-ASB-Pay**), it can be seen that bidding true cycling degradation costs will bring the ESSs the most profit ($2442.6). The ESSs' profit is determined by their social welfare contribution instead of the intertemporal price spreads, so the diminishing price spreads will not distort their incentives to behave honestly. Additionally, the ESSs will not lie about their ending SOC valuation either since it will decrease their profit.

The *incentive compatibility* will contribute to the social welfare. In **case VCG-ASB-Pay**, the rational ESSs will submit true parameters, and the ISO can dispatch the resources most efficiently. The corresponding social welfare contribution of the ESSs is $2442.6 (Table 2). However, in **case LMP-Pay**, the ISO will be misled by the false biddings and the clearing result will be suboptimal. For example, when the ESSs overstate their cycling costs by 40%, their social welfare contribution will decline by 12.2% to $2145.0. Similarly, in **case PAB-Pay**, the ESSs will gain more profit when overstating their costs by 60%, but the social welfare contribution will decline by 23.5% to $1867.7.

5.5. Computational tractability and relative fairness

In **case VCG-ASB-Pay**, the average computational time in seven scenarios is 119.06 s, which is 85.8% less than **case VCG-Pay** (840.44 s). The proposed settlement rule eliminates the needs to calculate each agent's marginal contribution one by one, and performs proper approximations instead.

Besides, the proposed mechanism preserves classical VCG's advantage of being fair. The revenue shares and contribution shares of the ESSs are shown in Table 4. In the classical VCG mechanism (**case VCG-Pay**), the revenue is calculated based on an agent's contribution, so the revenue share exactly matches the contribution share. In out proposed mechanism (**case VCG-ASB-Pay)**, it can be seen that the revenue share of an individual ESS generally matches its contribution share. Even when an agent owns several ESSs, the revenue share and contribution share still generally matches. For example, if an agent owns ESS 1–4, its revenue share will be 45.47% and its contribution share will be 45.37%.

To conclude, the proposed settlement outstrips the LMP settlement and PAB settlement, in that it provides proper incentives for the ESSs to behave honestly. It settles the ESSs based on their market contributions and overcomes the weakness originating from the diminishing intertemporal price spreads. Besides, the proposed settlement also outstrips the classical VCG settlement, in that it saves considerable time while generally preserving fairness.

6. Discussion and conclusion

This paper proposes improvements on the current pool-based market mechanism to accommodate the integration of the ESSs. To reflect the coupled relationship between charging and discharging and help the ESSs suggest their variable costs, a novel bidding structure and a corresponding clearing model are proposed, where the ESSs submit their cycling degradation costs and ending SOC valuation. To provide better incentives, a modified settlement rule is designated for the ESSs, where the aggregated payment to the ESSs is determined by VCG mechanism and further allocated among the ESSs by using Asymmetric Nash Bargaining theory.

The proposed mechanism is scalable and flexible in practical market operation.

Firstly, it is scalable for its computational tractability. Though incorporating a novel bidding structure based on cycling mileages, the clearing optimization problem remains linear, thus tractable for large-scale practical electricity market. By combining VCG mechanism and Asymmetric Nash Bargaining theory, the settlement rule overcomes the weakness of the traditional VCG mechanism that the computation is complicated especially when the number of agents is large.

Secondly, it is flexible due to compatibility with the currently operated mechanism. The designated bidding structure and settlement rule can operate in parallel with the current mechanism. ESSs can better suggest their parameters and they are encouraged to submit true biddings. In the meantime, generators and consumers can still choose the current bidding structure and settlement rule. In short, out work brings new mechanism elements but does not make fundamental modifications to the existing elements, so it is easy to gain acceptance within market participants.

Numerical tests show the advantages of the proposed mechanism over the current mechanism. Under the proposed bidding rule, the ESSs can suggest their accurate operating parameters and the ISO will manage the SOC flexibly. This contributes to both the ESS profit and the social welfare. The settlement rule pays the ESSs according to their contributions, so the ESSs are incentivized to behave honestly. The ISO who collects true parameters can generate welfare-maximizing clearing results. Besides, the introduction of Asymmetric Nash Bargaining theory can save considerable time while preserving fairness of the allocation. We hope the research could provide useful reference for the market mechanism design for the ESSs.

CRediT authorship contribution statement

Xichen Fang: Conceptualization, Formal analysis, Writing – original draft. **Hongye Guo:** Conceptualization, Formal analysis, Writing – original draft. **Xian Zhang:** Writing – original draft, Formal analysis. **Xuanyuan Wang:** Writing – original draft, Formal analysis. **Qixin Chen:** Conceptualization, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

1. Budget deficit caused by the VCG payment

In this subsection we prove that the VCG payment to the ESSs will incur budget deficit.

Lemma1. *For any given* $\mathbf{c}_{-\mathbf{I}}$, $S_{-I}(\mathbf{c}_{-\mathbf{I}},\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS}})$ *is concave in* $\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS}}$ *over* $\mathscr{Q}$.

Proof: For any given $\mathbf{c}_{-\mathbf{I}}$, let us consider two cases with different ESS scheduling plans, $\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS},1}$ and $\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS},2}$. We denote the corresponding optimal solution to (2.1)-(2.2) as $(\mathbf{P}_{\mathbf{D}}^{\mathbf{D},1},\mathbf{P}_{\mathbf{J}}^{\mathbf{G},1},\delta^1)$ and $(\mathbf{P}_{\mathbf{D}}^{\mathbf{D},2},\mathbf{P}_{\mathbf{J}}^{\mathbf{G},2},\delta^2)$. Since (2.1)-(2.2) is a linear programming problem, $(\frac{\mathbf{P}_{\mathbf{D}}^{\mathbf{D},1}+\mathbf{P}_{\mathbf{D}}^{\mathbf{D},2}}{2},\frac{\mathbf{P}_{\mathbf{J}}^{\mathbf{G},1}+\mathbf{P}_{\mathbf{J}}^{\mathbf{G},2}}{2},\frac{\delta^1+\delta^2}{2})$ must be a feasible solution when the argument $\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS}}$ takes the value of $\frac{\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS},2}+\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS},2}}{2}$. However, it is not necessarily the optimal solution, so the following formula can be derived.

$$\begin{aligned}
&\frac{S_{-I}(\boldsymbol{c}_{-I},\boldsymbol{P}_I^{ESS,1})+S_{-I}(\boldsymbol{c}_{-I},\boldsymbol{P}_I^{ESS,2})}{2}\\
&=\sum_{d\in D,k,t}\left(c_{d,k,t}^{\mathrm{D}}\frac{P_{d,k,t}^{\mathrm{D},1}+P_{d,k,t}^{\mathrm{D},2}}{2}\right)-\sum_{j\in J,k,t}\left(c_{j,k,t}^{\mathrm{G}}\frac{P_{j,k,t}^{\mathrm{G},1}+P_{j,k,t}^{\mathrm{G},2}}{2}\right)\\
&\leqslant S_{-I}(\boldsymbol{c}_{-I},\frac{\boldsymbol{P}_I^{ESS,1}+\boldsymbol{P}_I^{ESS,2}}{2})
\end{aligned}\tag{5.1}$$

The formula implies that S_{-I} is concave in $\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS}}$. ∎

Theorem 1. *The VCG payment made to the ESSs is no less than the LMP payment.*

Proof. Equation (2.1)-(2.2) *is a linear programming problem, so the feasible region should be convex. Since* $S_{-I}(\mathbf{c}_{-\mathbf{I}}^0,\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS},*})$ *and* $S_{-I}(\mathbf{c}_{-\mathbf{I}}^0,0)$ *is well defined, for any* $x\in[0,1]$, $S_{-I}\left(\mathbf{c}_{-\mathbf{I}}^0,\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS},*}-x\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS},*}\right)$ *is well defined.*

Due to the concavity of S_{-I} in $\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS}}$, we have

$$(1-x)S_{-I}(\boldsymbol{c}_{-I}^{\mathbf{0}},\boldsymbol{P}_I^{ESS,*})+xS_{-I}(\boldsymbol{c}_{-I}^{\mathbf{0}},\mathbf{0})\leqslant S_{-I}(\boldsymbol{c}_{-I}^{\mathbf{0}},\boldsymbol{P}_I^{ESS,*}-x\boldsymbol{P}_I^{ESS,*})\tag{5.2}$$

Let $x\to 0^+$ we have

$$S_{-I}(\boldsymbol{c}_{-I}^{\mathbf{0}},\boldsymbol{P}_I^{ESS,*})-S_{-I}(\boldsymbol{c}_{-I}^{\mathbf{0}},\mathbf{0})\geqslant \lim_{x\to 0^+}\frac{S_{-I}(\boldsymbol{c}_{-I}^{\mathbf{0}},\boldsymbol{P}_I^{ESS,*})-S_{-I}(\boldsymbol{c}_{-I}^{\mathbf{0}},\boldsymbol{P}_I^{ESS,*}-x\boldsymbol{P}_I^{ESS,*})}{x}\tag{5.3}$$

The left side of (5.3) is the VCG payment to the ESSs according to the definition in equation (2.3).

Let us denote $P_I^{ESS,*}=(P_{i,t}^{\mathrm{ESS,ch},*}(\forall i,t),P_{i,t}^{\mathrm{ESS,dis},*}(\forall i,t))$. According to reference [54], the LMP could be represented as the derivative of the welfare with respect to the net power injection to the corresponding node.

$$\begin{aligned}
&\frac{\partial S_{-I}^{+}(\boldsymbol{c}_{-I}^{\mathbf{0}},\boldsymbol{P}_I^{ESS,*})}{\partial\boldsymbol{P}_I^{ESS,*}}\\
&=\left(\frac{\partial S_{-I}^{+}(\boldsymbol{c}_{-I}^{\mathbf{0}},\boldsymbol{P}_I^{ESS,*})}{\partial P_{i,t}^{\mathrm{ESS,ch},*}}(\forall i,t),\frac{\partial S_{-I}^{+}(\boldsymbol{c}_{-I}^{\mathbf{0}},\boldsymbol{P}_I^{ESS,*})}{\partial P_{i,t}^{\mathrm{ESS,dis},*}}(\forall i,t)\right)\\
&=(-\lambda_{n(i\in\Omega_n^{\mathrm{ESS}}),t}(\forall i,t),\lambda_{n(i\in\Omega_n^{\mathrm{ESS}}),t}(\forall i,t))
\end{aligned}\tag{5.4}$$

Therefore, the right side of (5.3) could be reformulated as:

$$\begin{aligned}
&\lim_{x\to0^+}\frac{S_{-I}(c_{-I}^0,P_I^{ESS*})-S_{-I}(c_{-I}^0,P_I^{ESS,*}-xP_I^{ESS,*})}{x}\\
&=\lim_{x\to0^+}\frac{(\frac{\partial S_{-I}^{+}(c_{-I}^0,P_I^{ESS,*})}{\partial P_{i,t}^{ESS,ch,*}}(\forall i,t),\frac{\partial S_{-I}^{+}(c_{-I}^0,P_I^{ESS,*})}{\partial P_{i,t}^{ESS,dis,*}}(\forall i,t))\hat{A}\cdot(xP_I^{ESS,*})}{x}\\
&=\lim_{x\to0}\frac{(-\lambda_{n(i\in\Omega_n^{ESS}),t}(\forall i,t),\lambda_{n(i\in\Omega_n^{ESS}),t}(\forall i,t))\cdot(xP_I^{ESS,*})}{x}\\
&=\sum_{i,t}\lambda_{n(i\in\Omega_n^{ESS}),t}(P_{i,t}^{ESS,dis,*}-P_{i,t}^{ESS,ch,*})\\
&=R^{ESS,LMP}
\end{aligned}\tag{5.5}$$

Therefore, we can derive

$$R^{\mathrm{ESS,VCG}}\geqslant R^{\mathrm{ESS,LMP}}\tag{5.6}$$

Theorem 1 can be intuitively explained as follows: The LMP, in essence, calculates the welfare disturbance when there is a disturbance in the net nodal power injection. The VCG mechanism, in essence, calculates the welfare change when all the net power injection from an agent is excluded. Mathematically, the LMP payment is based on the tangent slope of function S_{-I} at point $\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS},*}$, and the VCG payment is based on the secant slope of the function S_{-I} between two points 0 and $\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS},*}$. Since $S_{-I}(\mathbf{b}_{-I},\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS}})$ is concave in the argument $\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS}}$, the secant slope is bigger than the tangent slope, and this makes the VCG payment higher than the LMP payment. A simplified demonstration is illustrated in Fig. 6.

The VCG settlement to the ESSs is more than the LMP settlement, so deficit will be incurred in the current budget-balanced LMP-settled market.

2. *Proof of Individual Rationality*

In this subsection we prove that the ESS reward under the LMP mechanism can always cover the costs of the ESSs.

Proof: In optimization problem (1.1)-(1.15), we derive the Lagrange conditions with regard to $P_{i,k,t}^{\mathrm{ESS,ch}}$ and $P_{i,k,t}^{\mathrm{ESS,dis}}$:

$$\frac{\partial L}{\partial P_{i,k,t}^{\mathrm{ESS,ch}}} = -c_{i,k}^{\mathrm{ESS,ch}} - \lambda_{n(i\in\Omega_n^{\mathrm{ESS}}),t} + \underline{\mu_{i,k,t}^{\mathrm{ESS,ch}}} - \overline{\mu_{i,k,t}^{\mathrm{ESS,ch}}} + \nu_i^{\mathrm{ESS}}\left(\eta_i^{\mathrm{ESS,ch}} - \sum_{\tau\geqslant t}\left(\overline{\tau_{i,\tau}^{\mathrm{ESS}}} - \underline{\tau_{i,\tau}^{\mathrm{ESS}}}\right)\right) = 0, \quad \forall i\in I,k,t \tag{6.1}$$

$$\frac{\partial L}{\partial P_{i,k,t}^{\mathrm{ESS,dis}}} = -c_{i,k}^{\mathrm{ESS,dis}} + \lambda_{n(i\in\Omega_n^{\mathrm{ESS}}),t} + \underline{\mu_{i,k,t}^{\mathrm{ESS,dis}}} - \overline{\mu_{i,k,t}^{\mathrm{ESS,dis}}} + \frac{\sum_{\tau\geqslant t}\left(\overline{\tau_{i,\tau}^{\mathrm{ESS}}} - \underline{\tau_{i,\tau}^{\mathrm{ESS}}}\right) - \nu_i^{\mathrm{ESS}}}{\eta_i^{\mathrm{ESS,dis}}} = 0, \forall i\in I,k,t \tag{6.2}$$

By substituting (6.1)(6.2) into equation (3.4), we have

$$R_i^{\mathrm{ESS,LMP}} = \sum_{k,t}\left(c_{i,k}^{\mathrm{ESS,ch}}P_{i,k,t}^{\mathrm{ESS,ch}} + c_{i,k}^{\mathrm{ESS,dis}}P_{i,k,t}^{\mathrm{ESS,dis}}\right) + \Lambda \tag{6.3}$$

$$\begin{aligned}
\Lambda &= \left\{\sum_{k,t}\left[\begin{array}{c}\overline{\mu_{i,k,t}^{\mathrm{ESS,dis}}}P_{i,k,t}^{\mathrm{ESS,dis}} - \underline{\mu_{i,k,t}^{\mathrm{ESS,dis}}}P_{i,k,t}^{\mathrm{ESS,dis}} + \overline{\mu_{i,k,t}^{\mathrm{ESS,ch}}}P_{i,t}^{\mathrm{ESS,ch}} - \underline{\mu_{i,k,t}^{\mathrm{ESS,ch}}}P_{i,t}^{\mathrm{ESS,ch}} \\ +\nu_i^{\mathrm{ESS}}E_i^{\mathrm{ESS,cap}}\left(\mathrm{SOC}_{i,0}^{\mathrm{ESS}} - \mathrm{SOC}_{i,T}^{\mathrm{ESS}}\right) + \left(\overline{\tau_{i,t}^{\mathrm{ESS}}} - \underline{\tau_{i,t}^{\mathrm{ESS}}}\right)E_i^{\mathrm{ESS,cap}}\left(\mathrm{SOC}_{i,t}^{\mathrm{ESS}} - \mathrm{SOC}_{i,0}^{\mathrm{ESS}}\right)\end{array}\right]\right\} \\
&= \sum_{k,t}\left[\begin{array}{c}\overline{\mu_{i,k,t}^{\mathrm{ESS,dis}}P_{i,k}^{\mathrm{ESS,dis}}} - \underline{\mu_{i,k,t}^{\mathrm{ESS,dis}}P_{i,k}^{\mathrm{ESS,dis}}} + \overline{\mu_{i,k,t}^{\mathrm{ESS,ch}}P_{i,k}^{\mathrm{ESS,ch}}} - \underline{\mu_{i,k,t}^{\mathrm{ESS,ch}}P_{i,k}^{\mathrm{ESS,ch}}} \\ +\overline{\tau_{i,t}^{\mathrm{ESS}}}E_i^{\mathrm{ESS,cap}}\left(\overline{\mathrm{SOC}_i^{\mathrm{ESS}}} - \mathrm{SOC}_{i,0}^{\mathrm{ESS}}\right) + \underline{\tau_{i,t}^{\mathrm{ESS}}}E_i^{\mathrm{ESS,cap}}\left(\mathrm{SOC}_{i,0}^{\mathrm{ESS}} - \underline{\mathrm{SOC}_i^{\mathrm{ESS}}}\right) \\ +\nu_i^{\mathrm{ESS}}E_i^{\mathrm{ESS,cap}}\left(\mathrm{SOC}_{i,0}^{\mathrm{ESS}} - \mathrm{SOC}_{i,T}^{\mathrm{ESS}}\right)\end{array}\right] \\
&\geqslant \nu_i^{\mathrm{ESS}}E_i^{\mathrm{ESS,cap}}\left(\mathrm{SOC}_{i,0}^{\mathrm{ESS}} - \mathrm{SOC}_{i,T}^{\mathrm{ESS}}\right)
\end{aligned} \tag{6.4}$$

The first equation can be derived with easy reformulation. The second equation can be derived from the complementary slackness conditions of the problem (1). From the complementary slackness condition of equation (1.4), we can derive the first and the second item in the right side of the equation. Similarly, we can derive the third, the fourth, the fifth and the sixth items by (1.5) and (1.7). The fourth equation holds for the non-negativity of the dual variables.

By deriving the Lagrange condition with regard to $\mathrm{SOC}_{i,k}^{\mathrm{ESS,end}}$, we could get

$$\frac{\partial L}{\partial \mathrm{SOC}_{i,k}^{\mathrm{ESS,end}}} = \left\{c_{i,k}^{\mathrm{ESS,SOCend}} - E_i^{\mathrm{ESS,cap}}\nu_i^{\mathrm{ESS}} + E_i^{\mathrm{ESS,cap}}\left(\underline{\mu_{i,k}^{\mathrm{ESS,SOCend}}} - \overline{\mu_{i,k}^{\mathrm{ESS,SOCend}}}\right)\right\} = 0, \forall i\in I,k \tag{6.5}$$

After proper manipulation, we could get the following formula. By using (1.11), we could get the second line. By substituting (6.5) into the equation, we get the third line. The usage of complementary slackness of problem and the non-negativity of dual variables can help us get the fourth line and fifth line.

$$\begin{aligned}
\Lambda &\geqslant \nu_i^{\mathrm{ESS,end}}E_i^{\mathrm{ESS,cap}}\left(\mathrm{SOC}_{i,0}^{\mathrm{ESS}} - \mathrm{SOC}_{i,T}^{\mathrm{ESS}}\right) \\
&= \sum_k \mathrm{SOC}_{i,k}^{\mathrm{ESS,end}}\nu_i^{\mathrm{ESS,end}}E_i^{\mathrm{ESS,cap}} \\
&= \sum_k\left\{-c_{i,k}^{\mathrm{ESS,SOCend}}\mathrm{SOC}_{i,k}^{\mathrm{ESS,end}} + \mathrm{SOC}_{i,k}^{\mathrm{ESS,end}}E_i^{\mathrm{ESS,cap}}\left(\overline{\mu_{i,k}^{\mathrm{ESS,SOCend}}} - \underline{\mu_{i,k}^{\mathrm{ESS,SOCend}}}\right)\right\} \\
&= \sum_k\left\{-c_{i,k}^{\mathrm{ESS,SOCend}}\mathrm{SOC}_{i,k}^{\mathrm{ESS,end}} - \underline{\mathrm{SOC}_{i,k}^{\mathrm{ESS,end}}}E_i^{\mathrm{ESS,cap}}\underline{\mu_{i,k}^{\mathrm{ESS,SOCend}}} + \overline{\mathrm{SOC}_{i,k}^{\mathrm{ESS,end}}}E_i^{\mathrm{ESS,cap}}\overline{\mu_{i,k}^{\mathrm{ESS,SOCend}}}\right\} \\
&\geqslant -\sum_k c_{i,k}^{\mathrm{ESS,SOCend}}\mathrm{SOC}_{i,k}^{\mathrm{ESS,end}}
\end{aligned} \tag{6.6}$$

Therefore, we have

$$R_i^{\mathrm{ESS,LMP}} \geqslant \sum_{k,t}\left(c_{i,k}^{\mathrm{ESS,ch}}P_{i,k,t}^{\mathrm{ESS,ch}} + c_{i,k}^{\mathrm{ESS,dis}}P_{i,k,t}^{\mathrm{ESS,dis}}\right) - \sum_k c_{i,k}^{\mathrm{ESS,SOCend}}\mathrm{SOC}_{i,k}^{\mathrm{ESS,end}} = C_i^{\mathrm{ESS}} \tag{6.7}$$

Equation (6.7) means that the revenue collected under the LMP mechanism can always cover ESS total cost C_i^{ESS}, including degradation costs and value changes in ending stored energy.

3. *Proof of incentive compatibility for a monopolistic ESS agent*

In this subsection we prove that the VCG payment satisfies incentive compatibility for a monopolistic ESS agent.

The profit of the monopolistic agent can be expressed in equation (7). $\pi^{\mathrm{ESS,VCG}}(\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS}})$ represents the profit. $C^{\mathrm{ESS,VCG}}(\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS}})$ represents the cycling degradation costs and value change in stored energy of the ESSs when their scheduling plan is $\mathbf{P}_{\mathbf{I}}^{\mathbf{ESS}}$. $S_{-I}(\mathbf{c}_{-\mathbf{I}}^{0}, \mathbf{P}_{\mathbf{I}}^{\mathbf{ESS}})$ represents the welfare of other market participants excluding the agent, and $S(\mathbf{c}^0_{-\mathbf{I}}, \mathbf{P}^{\mathbf{ESS}}_{\mathbf{I}})$ represents the total welfare. S differs from S_{-I} in that it considers the costs and utilities of the ESSs.

$$\begin{aligned}\pi^{\mathrm{ESS,VCG}}(\boldsymbol{P}_I^{\boldsymbol{ESS}}) &= R^{\mathrm{ESS,VCG}}(\boldsymbol{P}_I^{\boldsymbol{ESS}}) - C^{\mathrm{ESS,VCG}}(\boldsymbol{P}_I^{\boldsymbol{ESS}}) \\ &= S_{-I}(\boldsymbol{c}^{\mathbf{0}}_{-I}, \boldsymbol{P}_I^{\boldsymbol{ESS}}) - S_{-I}(\boldsymbol{c}^{\mathbf{0}}_{-I}, \mathbf{0}) - C^{\mathrm{ESS,VCG}}(\boldsymbol{P}_I^{\boldsymbol{ESS}}) \\ &= S(\boldsymbol{c}^{\mathbf{0}}_{-I}, \boldsymbol{P}_I^{\boldsymbol{ESS}}) - S_{-I}(\boldsymbol{c}^{\mathbf{0}}_{-I}, \mathbf{0})\end{aligned} \tag{7}$$

The second equation holds according to the VCG payment calculation in equation (2.3). The third equation holds according to the definition differences between S_{-I} and S. The second item in the last equation is a constant, since it has nothing to do with the scheduling plan of the ESSs. Therefore, the monopolistic agent's profit is determined only by $S(\mathbf{c}^0_{-\mathbf{I}}, \mathbf{P}^{\mathbf{ESS}}_{\mathbf{I}})$.

The optimization problem (1) gets the welfare-maximizing scheduling $\mathbf{P}^{\mathbf{ESS},*}_{\mathbf{I}}$ when the monopolistic agent submits true parameters (including cycling degradation costs and ending SOC valuation). If the agent lies about the parameters, the corresponding scheduling will deviate from the welfare-maximizing one, and the reduction in social welfare $S(\mathbf{c}^0_{-\mathbf{I}}, \mathbf{P}^{\mathbf{ESS}}_{\mathbf{I}})$ will decrease the profit $\pi^{\mathrm{ESS,VCG}}(\mathbf{P}^{\mathbf{ESS}}_{\mathbf{I}})$. Therefore, the monopolistic agent have no incentive to submit false parameters.

The proof can also be used to verify *anti-collusion.* Since the aggregated payment to the ESSs is determined as a whole first no matter the number of ESS owners, there is no difference between the payment received by a monopolistic ESS owner and a grand coalition formed by several ESS owners. Therefore, the ESSs cannot collude to gain extra profit.

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